SULIT

UNIVERSITI TEKNOLOGI MALAYSIA
PEPERIKSAAN AKHIR SEMESTER I 2013/2014

KOD KURSUS : SCJ 3323

NAMA KURSUS : SENIBINA & REKABENTUK PERISIAN

TAHUN/PROGRAM :

MASA :

TARIKH :

TEMPAT :

ARAHAN KEPADA CALON (INSTRUCTIONS): :

SECTION A : 4 STRUCTURED QUESTIONS (60 MARKS)
SECTION B : 2 ESSAY QUESTIONS (40 MARKS)

PLEASE ANSWER ALL QUESTIONS IN THE BOOKLET.

(sila catatkan nama pensyarah dan seksyen di dalam kertas jawapan anda)

Nama	
No. K/P	
Tahun/Program	
Seksyen	
Nama Pensyarah	

KERTAS INI TERDIRI DARIPADA _5_ MUKA SURAT SAHAJA TIDAK TERMASUK MUKA SURAT INI.

SECTION A - Structured Questions **[60 marks]**

Instruction: Answer all questions. Write down your answer in the answer booklet.

Question 1 [15 marks]

a. During Detailed Design activity, following are the major tasks. Describe each of them. [5 marks]

i. Understanding the architecture and requirements
ii. Creating detailed designs
iii. Evaluating detailed designs
iv. Documenting software design
v. Monitoring and controlling implementation

b. What is the difference between Interface Segregation Principle (ISP) and Open-Closed Principle (OCP)? [2 marks]

c. What is the difference among these categories of design patterns? [3 marks]

i. Creational Design Pattern
ii. Structural Design Pattern
iii. Behavioral Design Pattern

d. Compare Architectural Patterns and Design Patterns. Provide TWO (2) differences for each. [4 marks].

Question 2 [15 marks]

a. Describe any FIVE (5) of the following design patterns. [5 marks].

i. Builder
ii. Prototype
iii. Observer
iv. Adaptor
v. Facade
vi. Singleton

b. Identify the types of patterns for each description below. Adaptor, Façade, Observer, Composite. [4 marks]

Description	Design Pattern
Which design pattern would you use in situation where if one object changes state, all its dependents are notified and updated automatically, using a one-to-many relationship.	i. _____
Which design pattern would you use to allow designers to compose (large) tree-like designs structures.	ii. _____
Which design pattern would you use to convert the interface of a class into	iii. _____

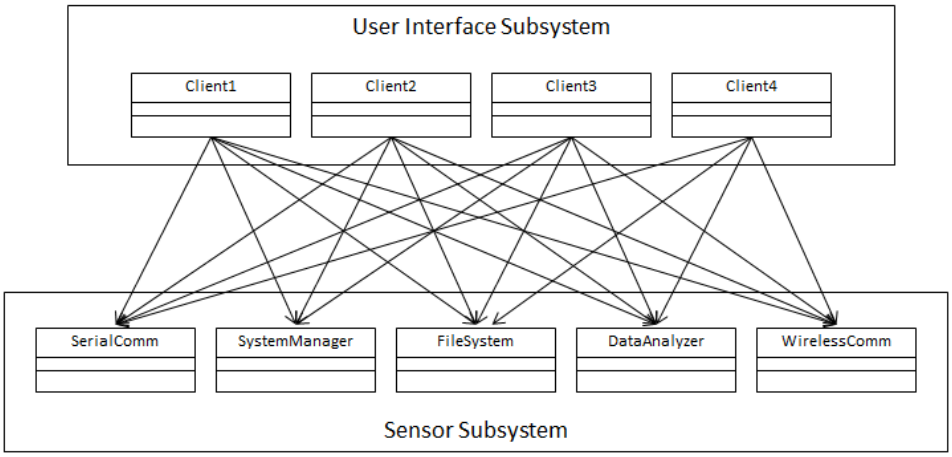
another interface clients expect?	
Which design pattern would you use to provide a unified interface to a set of interfaces in a subsystem	iv. _____

c. Identify the types of systems for each description below. Provide reason for your choice. **Hierarchical system, Data flow system, Interactive system, Data-centered system, Distributed system** [5 marks]

System Description	System	Reason
Systems oriented around the transport and transformation of a stream of data.		
Systems that serve as a centralized repository for data, while allowing clients to access and perform work on the data.		
Systems primarily involve interaction between several independent processing units connected via a network.		
Systems where components can be structured as a hierarchy (vertically and horizontally) to reflect different levels of abstraction and responsibility.		
Systems that serve users or user-centric systems.		

(5 marks)

d. Which Design Pattern is suitable to resolve the following complexity? **Builder, Prototype or Façade**. [2 marks]



Question 3 [15 marks]

a. What is the purpose of Behavioral Construction Design? Name three different techniques that can be used in the Behavioral Construction Design. [4 marks]

b. Programming Design Language (PDL) is sometimes used to design the internal function behavior. Briefly describe two benefits of using PDL. [2 marks]

c. Naming convention is one of the aspect that should be considered in Software Construction Styles. Give at least two examples of incomplete and inappropriate variable names, and suggest a better solution. [2 marks]

d. What is the difference between low-fidelity and high-fidelity prototypes? When is a suitable time to use low-fidelity prototype? [3 marks]

e. Gestalt Pronciples is one the common design methods for HCI. Describe at least four from six major Gestalt Principles that you know. [4 marks]

Question 4 [15 marks]

a. What is Software Design Management? [2 marks]

b. Based on the following information, calculate the Earned Value (EV), Cost Performance Index (CPI), Schedule Performance Index (SPI), Estimated Remaining Cost (ETC), and Estimated Total Cost (EAC). [5 marks]

- Number of task is a design project: 5 tasks.
- Estimated cost of each task: RM 2,000
- Total budget for the project (BAC): RM 10,000
- Estimated duration to complete each task: 1 month
- Estimated total planned duration for the project: 5 months.
- In the second month (i.e. current month), only one task have been fully completed at a cost of RM3,000.

c. Leadership is important in ensuring the success of a large scale design project. Personality dimension is used to categorize different personality traits in identifying an effective leader. Please write down a personality dimension for each of the following traits: [5 marks]

Description of personality traits	**Personality Dimension**
Individuals who are friendly, sociable, and compassionate.	i. _____
Individuals who are dominant (i.e., want to be in control), enjoy competition, and are willing to confront others, among other things.	ii. _____

Individuals who are in control of themselves, positive minded, and react well under pressure.	iii. _____
Individuals who are willing to work extra hours and make sacrifices to reach the assigned objectives.	iv. _____
Individuals who are flexible to change, open minded, and attempt enjoy trying new things.	v. _____

d. In order to make software engineering profession highly respected, software engineers must abide to the highest possible standards when developing software systems. Various principles of ethics may ensure that professionals in the software engineering field adhere to high-levels of ethical conduct. Briefly describe the following principles of ethics: [3 marks]

i. Public

ii. Product

iii. Management

SECTION B - Problem Solving / Essay Questions [40 marks]

Instruction: Answer all questions. Write down your answer in the answer booklet.

Question 1 [20 marks]

a. Draw a class diagram of the Design Pattern you adopted in your class project. Highlight which Design Pattern you used in that class diagram [10 marks]

b. Apply Adapter Design Pattern: [10 marks]

You have developed a Flying Car game, called SLOW CAR GAME. In your game, Car can do the following functions.

- Start ( )
- slowForward ( )
- slowReverse ( )
- slowFly ( )
- slowLand ( )
- stop ( )

You want to improve the Car to do the FAST features. For example, you want the Car to

- fastForward ( )
- fastReverse ( )
- fastFly ( )
- fastLand ( )

But, you think it is a time taking task to modify your code. So, you manage to find code from a friend who has already developed FAST features for his FAST CAR GAME, and he allows you to use his class diagram and code. Thus, you want to re-use his class diagram and code to improve your SLOW CAR GAME. For this purpose, it is appropriate to apply Adapter Design Pattern.

Your Task: Draw **a class diagram** so that SLOW CAR GAME can use the use the functions of FAST CAR GAME. Highlight the adapter class in your diagram.

Question 2 [20 marks]

Answer the questions based on the following system description:

Shopping application

The process flow for the shopping application may be described as follows:

- The service provider (shop) receives message (shopping list / shopping order by fax or e-mail) from final user (customer) specifying order (what and when to be delivered). The order details are recorded in the system. The initial status of order is set to 'received'.
- Shop then responsible for 'order picking' goods ordered as necessary. If the stock is available, the order status is set to 'ready for delivery'. If the stock is not available for some of the items, they will be marked with 'not available'. However, if the stock is not available for all items, the order status is set to 'not available'. The status of the order is set to rejected if the customer has a bad credit record with the shop.
- Goods packed ready for picking up from shop at time/date specified for delivery to customer. The order status is changed to 'ready for delivery'.
- The goods are then delivered to home or pick-up point. The order status is changed to 'despatched'.
- If the customer is not at home, the goods will be delivered to a pick-up point. The order status is updated to 'sent to pick-up point'. The arrangements for later pick-up/delivery by the customer is made manually over the phone or SMS.
- receive cash-on-delivery payment or proof of online payment. The payment details are recorded in the system via mobile device. The order status is changed to 'paid'.

a. Develop a flow-based model that represent the algorithmic viewpoint. [10m]

b. Draw a state-based model for an **`order`** object. [10m]